

Pipeline Blockers and Precondition Reference Manual

This document specifies the exact, codebase-level preconditions, predecessor dependencies, and credentials blocks that govern each step of our multimodal ingestion and reasoning pipeline.

Symmetrically, the orchestrator enforces these boundaries to prevent out-of-order execution, ensuring data consistency and rate safety across every processing step.

Pre-Flight & Environment Initialization Blockers

Step 0: checkin

- **Target Command:** `rag checkin <project-name>`
- **Blocker / Precondition:** None. This command can always be executed to bind or switch active workspaces.

Step 1: init (environment)

- **Target Command:** `rag init`
 - **Blocker / Precondition:** Requires Step 0 (`checkin`) to be complete.
 - *Violation Behavior:* If the active project pointer marker is missing, the command immediately blocks with: `[FAIL] Active project pointer not found. Please run 'rag checkin <project-name>' before initializing!` and terminates with `ExitCode.ENVIRONMENT_INCOMPLETE` (5).
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Document Ingestion & Classification Blockers

Step 2: ingest

- **Target Command:** `rag ingest`
- **Blocker / Precondition:** Requires Step 1 (`init`) to be complete.

- *Violation Behavior:* The orchestrator blocks execution unless the environment audit passed marker exists inside the project directory.

Step 3: define-layers

- **Target Command:** `rag define-layers`
- **Blocker / Precondition:** Requires Step 2 (`ingest`) to be complete.
 - *Violation Behavior:* The orchestrator blocks execution unless the document inventory exists and contains at least one registered technical PDF.

Step 4: propose-layers

- **Target Command:** `rag propose-layers`
- **Blocker / Precondition:** Requires Step 3 (`define-layers`) to be complete.
 - *Violation Behavior:* The orchestrator blocks execution unless the frozen layers ontology database file exists inside the workspace directory.

Visual Discovery & Verification Blockers (Pass 1)

Step 5: macro-images

- **Target Command:** `rag macro-images`
- **Blocker / Precondition:** Requires Step 4 (`propose-layers`) to be complete.
 - *Violation Behavior:* The orchestrator blocks execution unless document-level layer classifications are approved/saved to the inventory.

Step 6: verify-drafts

- **Target Command:** `rag verify-drafts`
 - **Blocker / Precondition:** Requires Step 5 (`macro-images`) to be complete.
 - *Violation Behavior:* The orchestrator blocks execution unless the drafting queue manifest (`crops/drafts.yaml`) physically exists on disk.
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Visual Subdivision & Verification Blockers (Pass 2)

Step 7: micro-images

- **Target Command:** `rag micro-images`
- **Blocker / Precondition:** Requires Step 5 (`macro-images`) to be complete. (Note: this is an explicit pipeline exception to bypass corrupt or stale macro-image drafts inside Step 6).
 - *Violation Behavior:* The orchestrator blocks execution if Step 5 has not successfully run to generate the drafts queue.

Step 8: verify-drafts --micro

- **Target Command:** `rag verify-drafts --micro`
- **Blocker / Precondition:** Requires Step 7 (`micro-images`) to be complete.
 - *Violation Behavior:* The orchestrator blocks execution unless the subdivision checklists exists and have been processed on disk.

Human Review & Annotation Blockers

Step 9: review (regions) (HITL Gate 3)

- **Target Command:** `rag review`
- **Blocker / Precondition:** Requires Step 5 (`macro-images`) to be complete. (Note: this is an explicit pipeline exception to allow the user to completely skip Pass 2 micro-subdivisions if they only wish to review and catalog macro-objects).
 - *Violation Behavior:* The orchestrator blocks execution if Step 5 has not successfully run to generate the drafts queue.

Step 10: caption

- **Target Command:** `rag caption`
 - **Blocker / Precondition:** Requires Step 9 (`review (regions)`) to be complete.
 - *Violation Behavior:* Symmetrically, the captioner refuses to execute on any region unless that region has been explicitly human-approved at Gate 3. If any region is un-reviewed or rejected, captioning for those items is bypassed.
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Assets Minting, Layer Assignment, & Publishing Blockers

Step 11: ingest --images

- **Target Command:** `rag ingest --images`
- **Blocker / Precondition:** Requires Step 10 (`caption`) to be complete.
 - *Violation Behavior:* Symmetrically, URN asset minting is completely refused if any approved region is missing a caption or vision prompt inside the database.

Step 12: assign-layers

- **Target Command:** `rag assign-layers`
- **Blocker / Precondition:** Requires Step 11 (`ingest --images`) to be complete.
 - *Violation Behavior:* The orchestrator blocks execution unless URN assets exist in the inventory database.

Step 13: review --assets (HITL Gate 4)

- **Target Command:** `rag review --assets`
- **Blocker / Precondition:** Requires Step 12 (`assign-layers`) to be complete.
 - *Violation Behavior:* The orchestrator blocks execution unless layer assignments have been run on the assets.

Indexing, Verification, & Q&A Blockers

Step 14: build-index

- **Target Command:** `rag build-index`
- **Blocker / Precondition:** Requires Step 13 (`review --assets`) to be complete.
 - *Violation Behavior:* Symmetrically, index compilation is blocked and refused if any asset is pending human review or lacks approval at Gate 4.

Step 15: verify

- **Target Command:** `rag verify`
- **Blocker / Precondition:** Requires Step 14 (`build-index`) to be complete.

- *Violation Behavior:* Symmetrically, deep verification is blocked if the required vector and keyword indexes are missing.

Step 16: export

- **Target Command:** `rag export`
- **Blocker / Precondition:** Requires Step 15 (`verify`) to be complete.
 - *Violation Behavior:* Strictly blocked unless the deep verification success marker `logs/.verify.ok` exists on disk. Symmetrically, if missing, the command raises a `VerificationIncompleteError` and exits with `ExitCode 1`.

Step 17: query (warmup)

- **Target Command:** `rag query`
- **Blocker / Precondition:** Requires Step 16 (`export`) to be complete.
 - *Violation Behavior:* Bypassed unless exported machine catalogs exist inside the workspace directory.

Step 18: chat

- **Target Command:** `rag chat`
- **Blocker / Precondition:** Requires Step 17 (`query (warmup)`) to be complete.
 - *Violation Behavior:* Bypassed unless the warmup query passed marker `logs/.warmup.ok` exists on disk.